

ComplexGitSync 0002.51– Python API

Nicolas Flipó

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1 Direct Python Object API

This chapter shows how to manipulate `ComplexGitSync` objects directly (without the CLI wrapper), including loading a `.gts` snapshot, reconstructing the runtime registry, mirroring identity into `GitTree/GitRepo`, and propagating a tag target.

For normal Multi-repo Sync usage, prefer the CLI lifecycle documented in the README and Getting Started guide. The reference test topology is `CGSill` (https://gitlab.com/CGS_test/CGSill1): `initialise -> pull -> checkout/add/commit/push -> freeze -> launch_release`.

The lower-level object/API lifecycle remains: `load(.cgs) -> .gts LOADED -> expand(.cgs|.gts) -> .gts PENDING -> validate(.cgs|.gts) -> .gts READY -> pull(.cgs|.gts) -> checkout(.gts) -> add -> git(tree,"commit",msg) -> git(tree,"push") -> git(tree,"tag",name) -> freeze`.

2 Programmatic Project Configuration

Project configuration does not require the CLI or an existing `.cgs` file. The public facade accepts authoring values non-interactively and delegates normalization, static validation, and optional serialization to `cgs_format.py`. It performs no Git or network operations.

```
from ComplexGitSync import ComplexGitSyncClient

client = ComplexGitSyncClient()
document = client.configure(
    "GX4G",
    ["codeberg:GX4G/GX4G"],
    output_path="GX4G.cgs",  # optional
)
reference_tree = document.to_git_tree()
```

The CLI commands `configure`, `create-cgs`, and `direct initialise -project/-repo` authoring are adapters over this method.

3 From Empty Registry to READY via `.gts`

```
from pathlib import Path

from ComplexGitSync import GitRepo, GitTree, RefKind, TreeLifecycleState
from ComplexGitSync.orchestration import GtsDocument,
    build_registry_from_gts_document
from ComplexGitSync.git_tree import WorkingGitTree

# 1) Empty registry lifecycle
empty_registry = WorkingGitTree()
assert empty_registry.recompute_tree_state() == TreeLifecycleState.
    UNLOADED

# 2) Load the CGSil1 runtime snapshot (.gts) and rebuild the dependency
    registry
cgshome = Path("../..")
gts_doc = GtsDocument.from_toml(cgshome / ".cgitsync/state/CGSil1.gts")
registry = build_registry_from_gts_document(gts_doc)
assert registry.lifecycle_state == TreeLifecycleState.READY

# 3) Rebuild GitTree/GitRepo identity objects from discovered registry
    entries
tree = GitTree()
for entry in registry.values():
    tree.add_repo(
        GitRepo(
            project_owner_name=entry.project_owner_name or "owner",
            project_name=entry.project_name or entry.name,
            gitprovider=entry.gitprovider,
            access_protocol=entry.access_protocol,
            commit_sha=entry.commit_sha,
        )
    )
```

4 Object-Level Tag Propagation

```
# Set the same tag target across all registry entries (parent -> leaves)
tree.propagate_tag(registry, "v1.2.3")
for entry in registry.values():
    assert entry.target_ref_kind == RefKind.TAG
    assert entry.target_ref_name == "v1.2.3"
```

This object-level flow is the same state model used by the CLI and `ComplexGitSyncClient`, but exposed directly for advanced automation and tests.

5 READY Methods in the Client API

The same lifecycle model is exposed by `ComplexGitSyncClient`. In practice, the workflow is:

- `load(...)` is the step-1 name for direct source loading.
- `expand(...)` is the canonical step-2 name; it runs nested `.cgs` discovery from parents to leaves (recursive).
- `validate(...)` is the canonical step-3 name; it accepts both `.cgs` and `.gts` sources.
- `initialise(...)` bootstraps a tree and must finish in `READY`; `clone(...)` is the direct clone entry point for a `.cgs` source; `pull(...)` resynchronises an existing tree.
- `git(tree, command, *args)` is the unified step-5 interface for "commit", "push", and "tag" operations. Each follows leaf-first ordering.
- `freeze(...)` emits the next persisted `.gts` state.

```
from pathlib import Path

from ComplexGitSync import ComplexGitSyncClient

client = ComplexGitSyncClient()
cgspath = Path("../..")
cgshome = cgspath / "CGSil1"

# 1. load the CGSil1 test-case .cgs -> .gts LOADED
client.load("../CGSil1.cgs")

# 2. expand(.cgs/.gts) -> .gts PENDING
print(client.expand("../CGSil1.cgs"))

# 3. validate(.cgs/.gts) -> .gts READY
client.validate("../CGSil1.cgs")

# 4. clone(.cgs/.gts)
# Same default as the CLI: output_path defaults to CGSPATH=../..
client.initialise("../CGSil1.cgs")

# Equivalent explicit form:
client.initialise("../CGSil1.cgs", output_path=cgspath)

# Recovery path for partial/stale clone state:
client.clean_initialise_cgs("../CGSil1.cgs", output_path=cgspath)

# Cleanup only:
```

```

client.purge_cgs("../CGSill1.cgs", output_path=cgspath)

# 5. READY methods - unified git interface
registry = client.get_dependency_registry()
# Pull uses ROOT -> PARENT -> LEAF; every repository in the tree is a
  plain
# independent clone (there are no gitlinks) and receives its own git
  pull.
# A .gts source must be an actual snapshot path (find the current one
  via
# RuntimeStateStore.latest_snapshot_for(), the same lookup the CLI uses
  when
# a command's snapshot argument is omitted) -- unlike a .cgs source,
  there is
# no fixed, predictable filename to hard-code here.
client.pull("../CGSill1.cgs")
# pull() fetches every branch of each remote, so checkout() can join a
  branch
# somebody else pushed: it starts the local branch from the remote-
  tracking
# ref when one exists, and only otherwise from the current HEAD.
client.checkout("feature/my-branch")
# Mutations use LEAF -> PARENT -> ROOT.
client.add()
client.git(registry, "commit", "feat: update CGSill1 CGS#1")
client.git(registry, "push")
client.git(registry, "tag", "v1.2.3")

# 6. freeze -> next .gts id
client.freeze("release-2026.05")

```

6 Landing a Branch: merge and refresh_private

`merge(project_branch, private=False, ff_only=False, no_ff=False)` merges a branch across the tree, leaf-first, and returns one `(repo_name, merged_ref)` pair per repository a merge moved.

The argument is always the **project's** branch name. Each repository resolves what that means for itself, so a private/local configuration repository merges `<its default_branch>_<project_branch>` rather than a branch of the project's name, which it does not have. `merge_plan(project_branch, private=False)` returns one `(repo_name, source_ref, status, conflicting_paths)` row per repository without merging anything — what the CLI's `-dry-run` prints. `status` is "merge", "already-on-it", "no-branch" or "conflicts", and the last one carries the files that block it.

Every repository in scope is checked before any is merged, so a conflict anywhere leaves the whole tree untouched. The check names the conflicting files, and counts binary files: a tracked PDF changed on both sides conflicts just as a source file does.

`merge_resolve(project_branch, private=False, ff_only=False, no_ff=False)` is the opt-in alternative. It merges one repository at a time, stops at the first that conflicts, and returns a `ResolveOutcome` naming what it merged, where it stopped, the files that stopped it, and the repositories it never reached. It gives up the all-or-nothing guarantee on purpose: that is the only way to reach the conflicted worktree a merge tool needs.

`open_merge_tool(repo_name)` then opens one repository's conflicts in a merge tool. It returns `None` once the tool has run, or the command to run by hand when no tool is available — a missing editor is a normal outcome, not an error. A `merge.tool` you configured always wins,

and your Git configuration is never written.

`refresh_private()` is the other direction, and what `pull -private` runs: it fetches and merges each private/local repository's base branch into the derived branch it is on, so a settings branch does not drift behind the project's while a feature branch is open.

```
client.merge("multi-branch")           # the project's own repos
client.merge("multi-branch", private=True) # ComplexGitSync_multi-
    branch                               # -> ComplexGitSync
client.merge("multi-branch", all_writable=True) # both halves, one
    pass                                 #
client.refresh_private()                 # the other way round
```

all_writable: the API half of `-all`. `add`, `commit`, `push`, `merge`, `merge_resolve` and `merge_plan` each take `all_writable=False` alongside `private=False`. Setting it runs the call across this project's own repositories *and* its writable configuration ones in a single pass. The two are mutually exclusive and passing both raises; leaving both unset keeps today's behaviour exactly. Read-only configuration repositories are never written to under any combination — the parameter maps to `RepoScope.WRITABLE`, which is narrower than `RepoScope.ALL`.

7 Targeting a Single Path: add and remove

`add(paths=None)` and `remove(paths)` accept one or more filesystem paths (each relative to `CGSHOME`, or absolute) instead of operating on the whole tree:

```
# Stage only these two files, each in the repo that owns it -- every
# other repo in the tree is untouched.
client.add(paths=["notes.md", "deps/leaf/src/main.py"])

# Remove a single tracked file: deleted from disk, removal staged.
client.remove(["deps/leaf/obsolete.txt"])
```

Both resolve each path to its owning repo through the same helper, `resolve_repo_for_path(tree, path)` (`git_tree.py`) — the deepest (most specific) repo whose `absolute_path` contains the resolved path wins, so a nested child is correctly preferred over its parent. A path outside every repo in the tree raises `GitSyncError` immediately, before anything is staged or removed; for `remove`, a path that resolves to a directory or does not exist raises the same way rather than partially applying. `add()` with no `paths` keeps today's exact behaviour: every repo staged in full, leaf-first.

`remove` is a plain `git rm` (deletes from disk, stages the removal) — unrelated to `GitRunner.rm_cached` (index-only, used internally by `import_submodules` to drop a gitlink while keeping the child's working tree); the two are not interchangeable.

8 .gitignore Lifecycle Sync and Commit Identity

`initialise_cgs`, `initialise_cgs_document`, `clean_initialise_cgs`, `clean_init`, `restart`, and `pull` (for `.cgs` sources) all accept the same four optional keyword arguments, mirroring the CLI's `-commit-gitignore`, `-force-gitignore-sync`, `-git-user-name`, and `-git-user-email` flags:

```
from ComplexGitSync import ComplexGitSyncClient, MasterConfig

client = ComplexGitSyncClient()
client.initialise_cgs(
```

```

    "../CGSill.cgs",
    output_path=cgspath,
    commit_gitignore=True,          # stage/commit/push each changed .
        gitignore
    force_gitignore_sync=False,    # pull-force fallback for a blocked
        repo
    git_user_name="cgitsync-bot",
    git_user_email="bot@example.com",
)

# The identity above is persisted to CGSHOME/.cgitsync/master.toml, so a
# later call on the same workspace picks it up without repeating it:
client.initialise_cgs("../CGSill.cgs", output_path=cgspath,
    commit_gitignore=True)
assert MasterConfig.resolve_identity(cgshome, client.git_runner) == (
    "cgitsync-bot",
    "bot@example.com",
)

```

All four keyword arguments default to **False/None**: by default, every repository with children (root, or any nested repository that itself has further nested children) is safely pulled and has its `.gitignore` updated, but nothing is staged, committed, or pushed, and the commit identity — were a commit ever made — would be whatever `git config user.name/user.email` already resolves to locally. `MasterConfig` (`master.py`) is the class behind the identity override: `configure()` sets it in memory for the current process, `persist(cgshome, ...)` writes it to `CGSHOME/.cgitsync/master.toml` (and also updates the in-memory value), `load(cgshome)` reads a previously persisted override back in, and `resolve_identity(repo_path, git_runner)` returns the `(user_name, user_email)` pair — or `(None, None)` to defer entirely to local git config. This file is workspace-local configuration, not part of the `.cgs/.gts` project spec, and is preserved (not deleted) by `purge/clean-init`.

9 Forcing a Clone Protocol

`initialise_cgs`, `initialise_cgs_document`, `clean_initialise_cgs`, `clean_init`, `bootstrap`, and `clone_cgs` additionally accept `force_access_protocol`, mirroring the CLI's `-force-protocol` flag on `initialise`, `bootstrap`, and `clean-init` (`restart/pull` do not accept it):

```

# Every repo this run clones -- root siblings and any child discovered
# later from a nested .cgs in a different repo -- is cloned over https,
# regardless of what its own .cgs entry says. No .cgs file is read
# differently or written.
client.initialise_cgs(
    "../CGSill.cgs",
    output_path=cgspath,
    force_access_protocol="https",
)

```

"ssh" or "https"; `None` (the default) leaves every entry's own `.cgs`-declared `access_protocol` untouched. The override lives only in memory for the duration of the call — it is applied at the single point every clone's remote URL is built (`ComplexGitSyncClient._build_remote_url`), so it reaches root siblings and nested-discovered children alike without `cgs_format.py` parsing, or any `.cgs` file on disk, ever being touched differently.

10 Repository Discovery: `discover_repos`

`discover_repos(root_dir=None, *, max_depth=5, output=None)` scans a directory for git repositories already checked out on disk and drafts a `.cgs` from what it finds — the entry point for adopting a project that exists but has no specification yet:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

# Read-only scan - nothing is cloned, fetched, or modified
report = client.discover_repos("/path/to/checkout")
for repo in report.repos:
    print(repo.relative_path, repo.identifier, repo.branch)

for warning in report.warnings:
    print("unresolved:", warning)

# Write the draft out (same call, with an output path)
report = client.discover_repos("/path/to/checkout", output="draft.cgs")
```

The method returns a `DiscoverReport` dataclass with fields `root` (the scanned directory), `repos` (a tuple of `DiscoveredRepo`), `cgs_entries` (authoring-form repo tables ready for `configure()`), `warnings`, and `project_name`. Each `DiscoveredRepo` carries `relative_path`, `absolute_path`, `remote_url`, `identifier`, `branch`, and `has_cgs`.

Repository addresses are resolved through the existing `parse_repo_id()` grammar; a repository with no origin or an unrecognised provider is reported in `warnings` and omitted from `cgs_entries` rather than guessed at. Passing `output` raises `GitSyncError` when nothing resolvable was found, so an empty draft is never written.

11 Submodule Migration: `import_submodules`

`import_submodules(repo_root, *, apply=False, recursive=False)` converts git submodules in a local repository to plain `ComplexGitSync` nested clones:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

# Dry run - inspect without changing anything
report = client.import_submodules("/path/to/repo")
for sub in report.submodules:
    print(sub.name, sub.path, sub.url)

# Apply - convert every level, root first
report = client.import_submodules(
    "/path/to/repo",
    apply=True,
    recursive=True,
)
print(report.converted)    # tuple of converted submodule names
```

The method returns an `ImportSubmodulesReport` dataclass with fields `submodules` (all entries found), `applied` (whether the conversion ran), `converted` (names of converted submodules), and `scan_root` (the repository the call was pointed at, which every reported path can be expressed from). When `apply=False` (the default), no files are modified. When `apply=True`, each submodule is removed from the index (`git rm -cached`), its stanza is removed from

`.gitmodules`, and the parent's `.gitignore` is updated to ignore the child path. The staged changes should then be reviewed and committed manually.

`recursive=True` also converts any submodule that has its own checked-out `.gitmodules`, at any depth. That walk follows the submodule graph declared by `repo_root`'s own `.gitmodules`, so a repository whose holder has already been converted is no longer reachable — which is why the conversion of a whole tree happens once, and last (see `init_from_submodules` below).

12 Adopting a Submodule Tree: `init_from_submodules`

`init_from_submodules(repo_root, *, cgs_path=None, max_depth=5, dry_run=False, force=False, force_access_protocol=None)` performs the whole adoption of a submodule-based checkout, in the one order that works:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()

# Report the plan without writing, cloning, or converting anything
plan = client.init_from_submodules("/work/project", dry_run=True)

# Adopt: discover -> write .cgs -> initialise -> convert every submodule
report = client.init_from_submodules("/work/project")
print(report.cgs_path)                # the .cgs used
print(report.tree.is_ready())          # True
print(report.import_report.converted)  # converted submodule names
```

Each step delegates to the method that already owns it (`discover_repos`, `initialise_cgs`, `import_submodules`); the value this method adds is the ordering and the preconditions. `initialise_cgs` adopts the root in place but deletes and re-clones every other repository from its remote, where the submodules are still declared — so converting first is undone for every non-root repository, and a repair pass cannot reach any deeper level once the root's `.gitmodules` is gone.

The method returns an `InitFromSubmodulesReport` with fields `root` (which is also `CGSHOME`), `discover` (the `DiscoverReport` from step 1), `cgs_path`, `cgs_written`, `import_report`, `tree`, and `dry_run`. Nothing is committed: the conversion touches every repository that held a submodule, so `branch`, `checkout`, `add`, and `commit` stay explicit, separate calls. A `GitSyncError` is raised before anything is written when the discovery resolved nothing, when the project name does not match the directory name (`CGSHOME` is `<parent>/<project-name>`), or when `repo_root` has no `.gitmodules` to convert (`force=True` overrides the last one).

13 Register Integrity: `verify`

`verify(cgshome, *, repair=False)` checks the hash-chained `.cgitsync/lgr` register under `cgshome` for tamper-evidence:

```
from ComplexGitSync.orchestre import ComplexGitSyncClient

client = ComplexGitSyncClient()
report = client.verify("/path/to/CGSHOME")

if report.is_clean:
    print("register is clean")
else:
    for seq, finding, detail in report.findings:
        print(seq, finding.name, detail)
```

```
# Repair a stale HEAD cache (never rewrites or deletes an entry)  
client.verify("/path/to/CGSHOME", repair=True)
```

Returns an `integrity.VerificationReport` (re-exported from the package root) whose `findings` list holds `(seq, Finding, detail)` tuples and whose `is_clean` property is `True` when the list is empty. A register with no entries yet is reported clean — this is the common case today, since no command writes to this per-entry format yet (the CLI’s `verify` command is the enforcement instrument; the migration of `SyncLedger`’s own writes to this format is a separate, future piece of work). `Finding` enumerates `BROKEN_LINK`, `BAD_ENTRY_HASH`, `SEQ_GAP`, `SEQ_DUPLICATE`, and `HEAD_STALE` for what this method actually checks (chain linkage and the HEAD cache); its remaining three members (`MISSING_STATE`, `ORPHAN_STATE`, `STATE_DIGEST_MISMATCH`) are reserved for a future store-level verification pass that cross-references entries against the actual `state(<hash>)_n/` directories on disk.

With `repair=True`, a stale HEAD cache is corrected in place. Register entries are never rewritten or deleted by this method, repair or not — a broken chain is reported, not silently healed.